

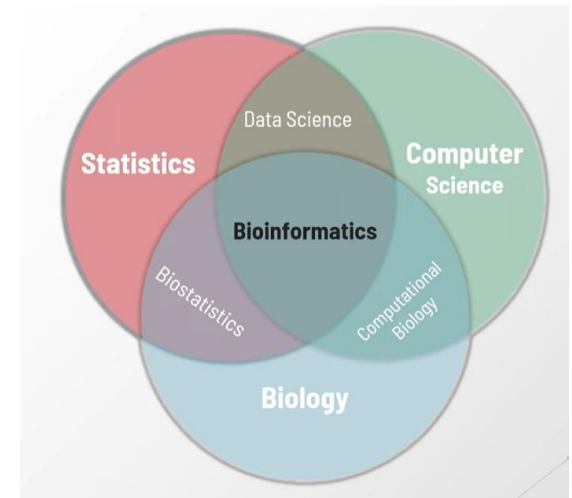
Bioinformatics



What is Bioinformatics?

- Bioinformatics is an interdisciplinary field of science in which biology, statistics/mathematics, computer science and information technology merge together to form a single discipline in order to analyze biological information using computers and statistical techniques.
- Biological experiments can generate large amounts of data (macromolecular sequences, structures, expression profiles, pathways and so on.)
- **Bioinformatics is about acquiring/ collecting, managing, analyzing and understanding those data.**

Also termed as biocomputing or computational Biology.



What is Bioinformatics?

Three important persons

Biologists:

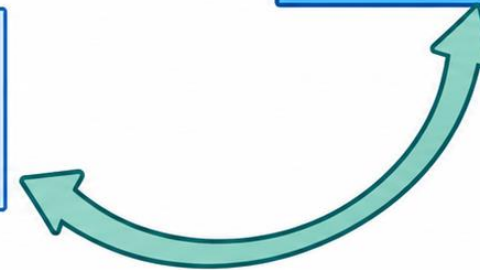
Collects Molecular data:
DNA & Protein sequence, gene expression etc

Computer scientists:

(+mathematicians, statisticians etc)
Develop tools, software, algorithms to store and analyze data

Bioinformaticians:

Study biological questions by analyzing
molecular data



What is Bioinformatics?

Explanation of the Diagram

1. Biologists

1. Collect molecular data.
2. Examples: DNA sequences, protein sequences, gene expression profiles, experimental results.
3. Provide the *raw biological information* that forms the basis for further analysis.

2. Computer Scientists (with Mathematicians & Statisticians)

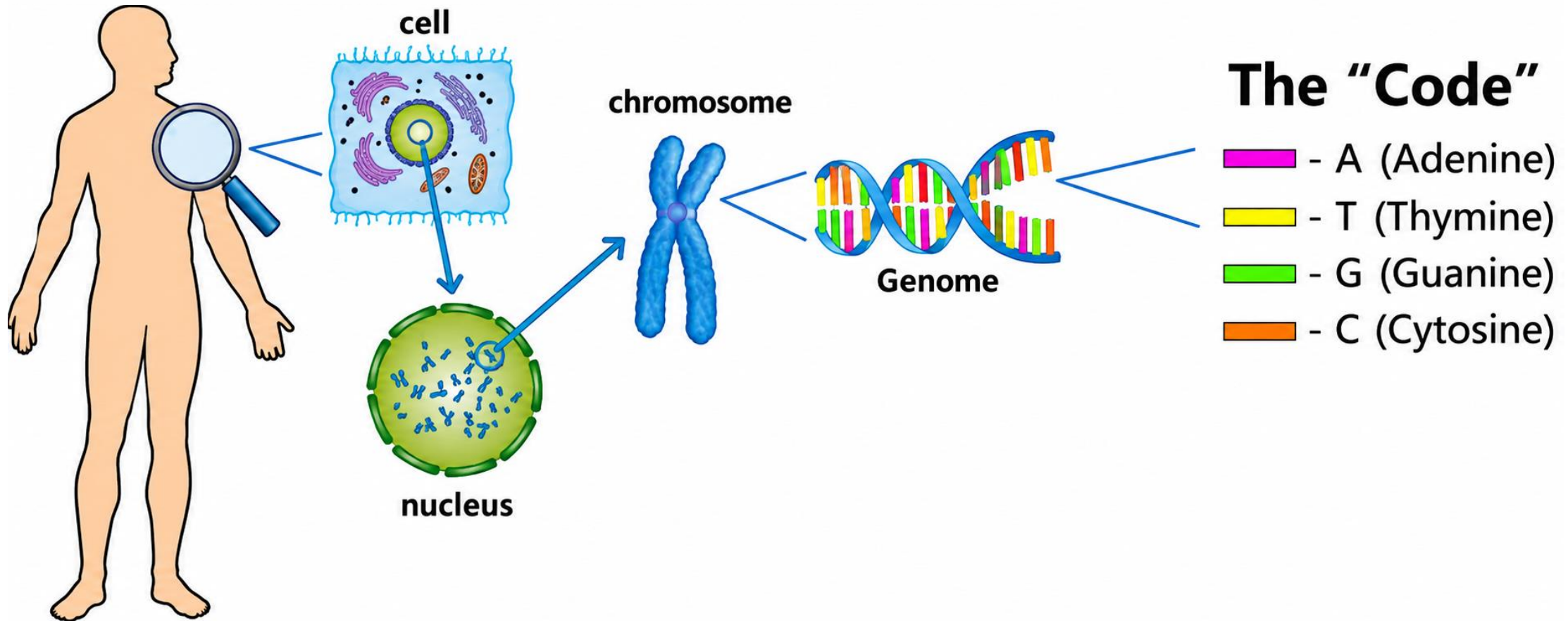
1. Develop computational tools, software, databases, and algorithms.
2. Enable the storage, processing, and large-scale analysis of massive biological datasets.

3. Bioinformaticians

1. Bridge the gap between biology and computer science.
2. Use computational methods to **study biological questions**—e.g., finding disease-causing genes, predicting protein structures, or analyzing evolutionary relationships.

Biological data

Bioinformatics is an interdisciplinary field that develops methods and tools for understanding biological data.



Biological Data

The **genome or DNA**, often called the “book of fate,” is a complete set of instructions written inside every cell of our body. Just as a book is made up of chapters, paragraphs, and sentences, our genome is composed of genes, sequences, and nucleotides. Within it lies the information that determines much of who we are—our gender, the height to which we may grow, the color of our eyes and skin, and even our resistance or susceptibility to certain diseases.

For centuries, this book remained closed to us. The sheer vastness of the human genome, with its billions of DNA letters, made it appear impossible to read and understand. However, recent decades, particularly with the completion of the **Human Genome Project** and advances in sequencing technologies, have allowed scientists to gradually “read” this book. What once seemed like an undecipherable mystery is now becoming a map that guides medicine, genetics, and our understanding of life itself.

Human Genome Project

❑ Human Genome Project (HGP)

- The Human Genome Project was an international scientific research effort that aimed to **map and understand all the genes of human beings**—collectively known as the *genome*. It officially began in **1990** and was completed in **2003**, two years ahead of schedule.

❑ The main goals of the HGP were to:

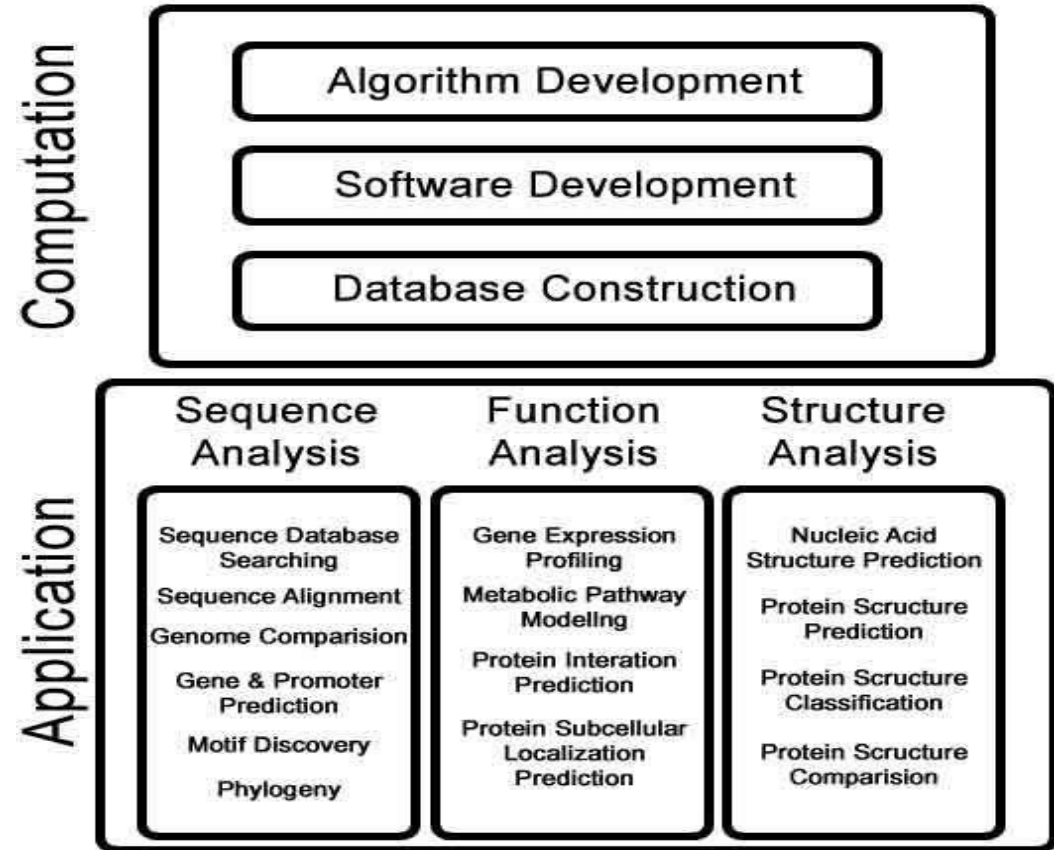
- **Identify all the genes** in human DNA (estimated to be about 20,000–25,000).
- **Determine the sequence** of the 3 billion DNA base pairs that make up the human genome.
- **Store this information** in databases to be accessible for research.
- **Improve tools** for data analysis.
- **Address ethical, legal, and social issues** arising from genome research.

Goals of Bioinformatics

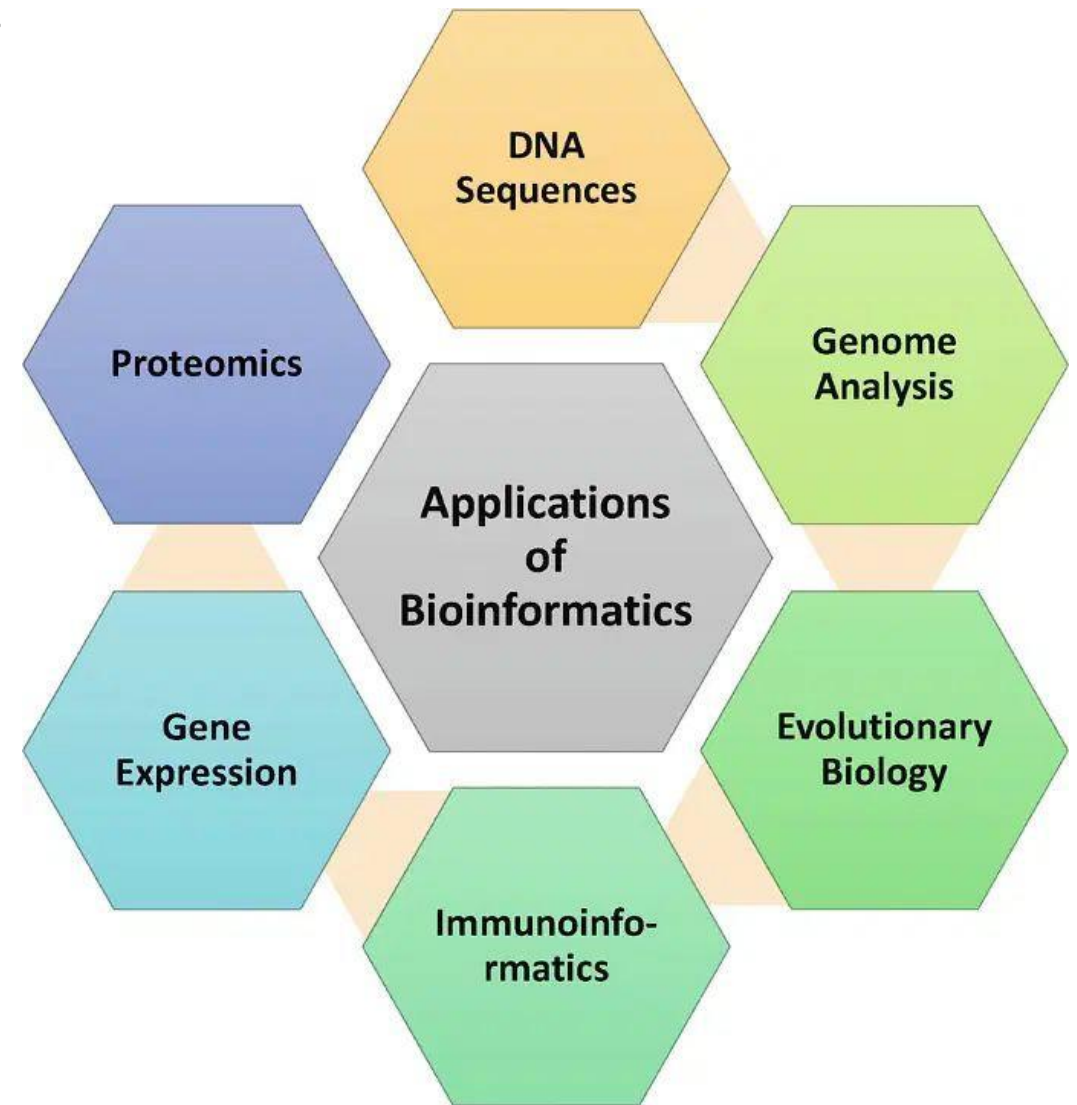
To construct and manage biological database.

To develop algorithm for different types of sequence analysis.

To develop softwares.



- Bioinformatics is about **analyzing huge biological datasets** to **find important facts**, which are then **processed and interpreted** to help in science and medicine. It's like turning unorganized raw data into meaningful insights.
- One major area is **3D image processing** and **modeling of living cells**, which enables scientists to visualize cellular structures in high resolution and simulate their behavior in virtual environments.
- The most important application of bioinformatics can be seen in the field of medicine, where its data is used to create antidotes for infectious and harmful diseases.



Uses of Bioinformatics

Bioinformatics is used in various fields, especially in biomedicine where it has various applications.

Drug discovery: By the use of structure-based drug design with Bioinformatics, scientists can come up with effective drugs for various illnesses and acute diseases.

Personalized medicine: By assessing the genetic structure of the patient and their respective medical history, personalized medicine can be curated which would prove to be much more effective.

Preventive medicines: This is primarily achieved by integrating data from bioinformatics, bioanalytics, and epidemiology. As the name suggests, preventive medicine focuses on interrupting the chain of disease transmission or mitigating acute conditions before their onset. By leveraging computational analyses, biological measurements, and population-level health data, preventive approaches can predict risks, design early interventions, and reduce the overall burden of disease.

Uses of Bioinformatics

Bioinformatics is used in various fields, especially in biomedicine where it has various applications.

Gene therapy: Gene therapy is the process where defective genes are replaced by new ones in the gene structure of a living organism. Since each organisms gene structure is very different, huge data may be involved to produce an accurate replacement.

Why is Bioinformatics important for CSE students?

- **Why *you*, as a CSE student, should study Bioinformatics?**

1. **It's CS applied to real, high-impact problems**

Bioinformatics uses:

- Algorithms & data structures
- Databases
- Machine learning
- Big data & cloud computing

...but instead of toy problems, you solve **real problems** like:

- Disease detection
- Drug discovery
- Genome analysis

Why is Bioinformatics important for CSE students?

2. Biology today is a data science problem

- Modern biology generates **terabytes of data** (DNA sequences, RNA-seq, protein data).
- Biologists **cannot analyze this alone**
- They need CSE people to design:
 - Efficient algorithms
 - Fast search techniques
 - Scalable systems
- Think of DNA as a *very long string* → classic CS problem.

Why is Bioinformatics important for CSE students?

3. Strong career advantage (interdisciplinary edge)

With bioinformatics, a CSE student can work as:

- Bioinformatics Engineer
- Computational Biologist
- Healthcare Data Scientist
- AI Engineer in biotech / pharma

These roles are:

- High demand
- Research-oriented
- Well-paid globally

Why is Bioinformatics important for CSE students?

4. Perfect fit if you like ML, AI, and Data Science

Bioinformatics is one of the **best application areas** for:

- Machine Learning
- Deep Learning
- Pattern recognition

Examples:

- Predicting diseases from gene data
- Cancer classification using gene expression
- Protein structure prediction (AI + biology)



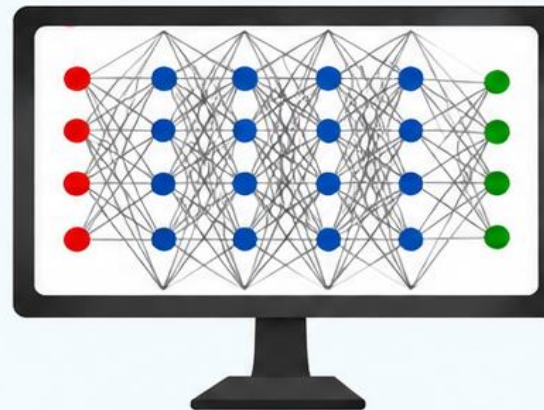
Traditional Bioinformatics Workflow



AGGAGTCAAATATCATGCGCAT

Manual processing and interpreting data is complex and lengthy

AI/ML Bioinformatics Workflow



AAATATCATGGGCGAT
TCAAATATCATGCG
AAATATCATGCGCAT
TCAAATATCATGCGCAT
AGTCAAATATCATGGCG
CAAATATCATGCGCAT

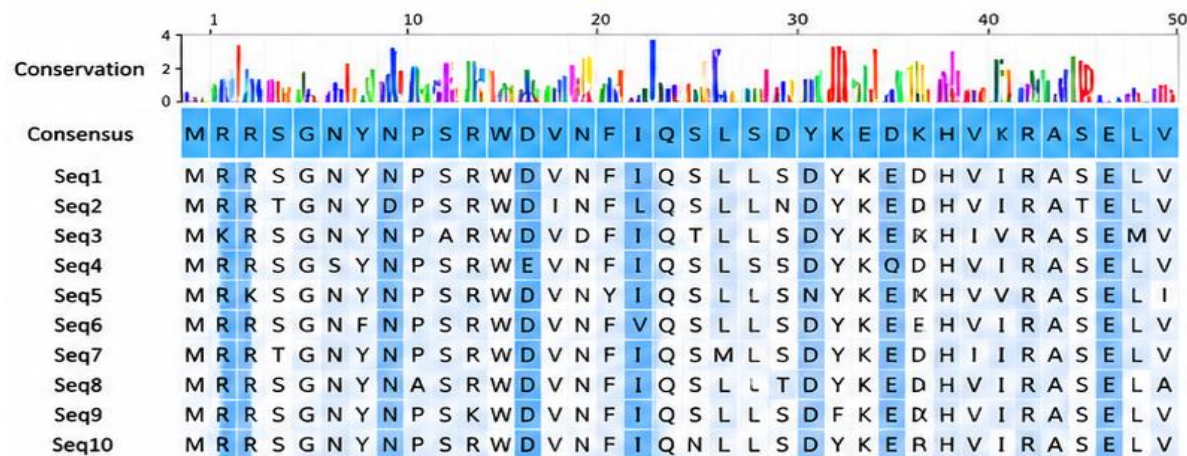
Automated AI/ML based analysis is faster and resourceful

Sequence

MR RSGNYNPSRWDVNF IQSL LSDYK EDKHVIRASELV



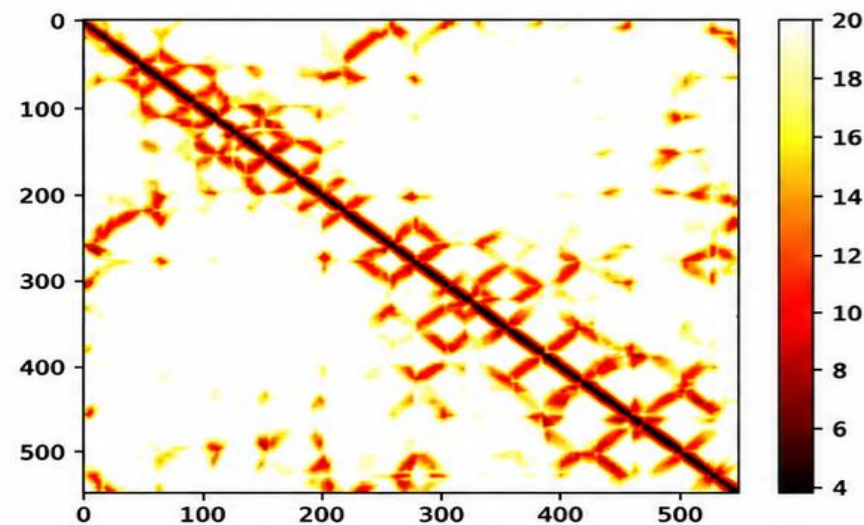
Multiple sequence alignment



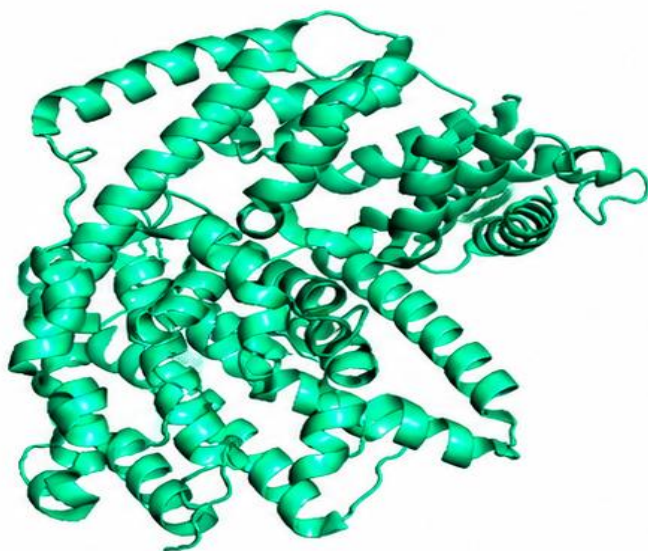
Deep learning



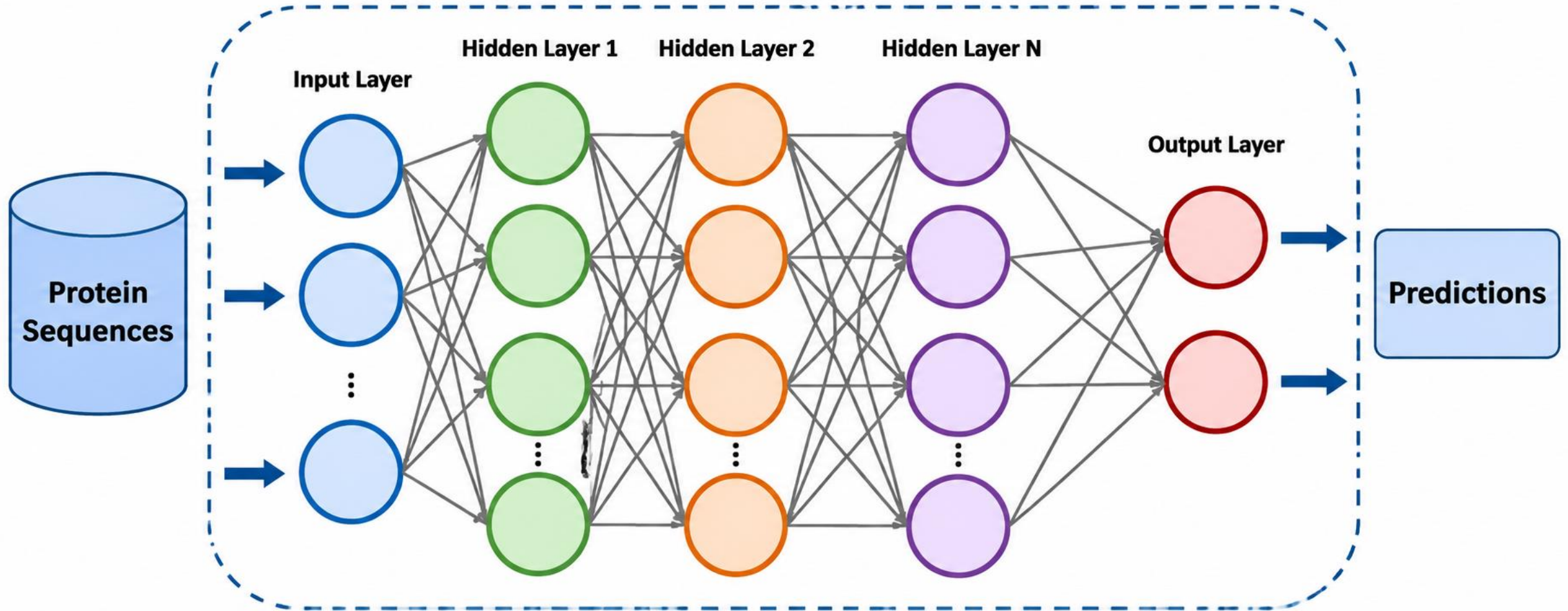
Distance prediction



3D structure



Feature Extraction & Classification



Why is Bioinformatics important for CSE students?

5. Research & higher studies advantage

If you plan:

- **MS / PhD**
- Research publications
- International scholarships

Bioinformatics makes your profile **stand out**, because:

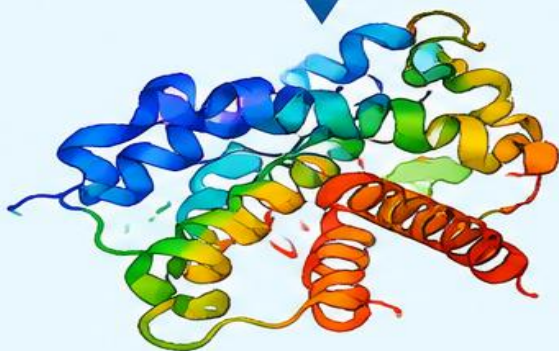
- It shows interdisciplinary thinking
- Many funded projects are in bioinformatics + AI

AI

...ANRKVSLQRYMAKE
REKRKDRKFSKYSYAA
KKEKRKEKS...

Protein sequence

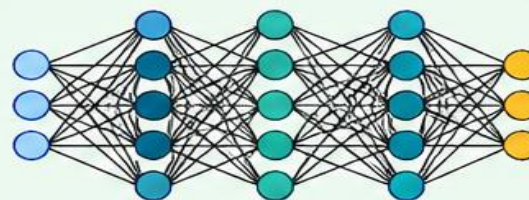
- AlphaFold series
- RoseTTAFold
- Language model-based tools



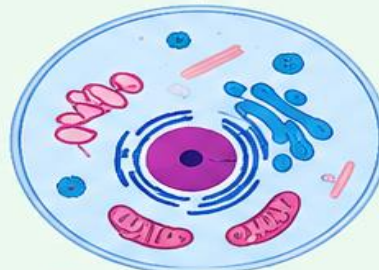
Protein
structure prediction

T	A	C	G	G	A	C	T	C	A	T
A	T	G	C	C	T	G	A	G	T	A

Sequence embedding



Neural networks



Function class

Protein function prediction
and target validation

Training data set

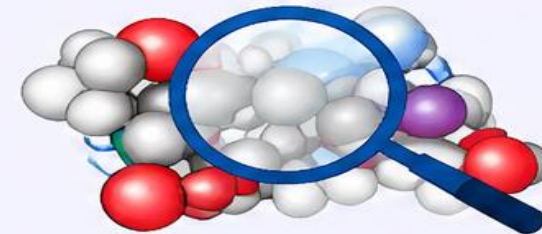
Test data set



Molecular descriptors



AI Model



Drug discovery

Data Collection

1

Data Types



Genomic Data

- Nucleotide sequences
- Gene expression
- Variants (SNP, CNV)
- Whole genome data



Proteomic Data

- Protein sequences
- Protein structures
- Protein-protein interactions
- Mass spectrometry data



Clinical Data

- Patient records
- Clinical phenotypes
- Laboratory results
- Treatment response



Metabolomic Data

- Metabolite profiles
- Small molecule data
- Metabolic pathways
- Spectrometry data

Data Sources



Public Databases

- NCBI, Ensembl
- UniProt
- PDB, KEGG
- GEO, SRA



Laboratory Experiments

- Sequencing
- Mass spectrometry
- Microarrays
- Clinical trials



Clinical and Medical Data

- Hospitals
- Electronic health records
- Biobanks
- Cohort studies



Scientific Literature

- Research articles
- Patents
- Text mining
- Knowledge graphs

Data Preprocessing

2



Raw Data



Data Cleaning

- Remove noise
- Handle missing data
- Correct errors



Data Integration

- Combine multi-omics data
- Merge heterogeneous data
- Standardize formats



Data Transformation

- Normalization
- Feature engineering
- Dimensionality reduction



Feature Selection

- Select relevant features
- Remove redundant data
- Reduce complexity



Data Balancing

- Handle class imbalance
- Resampling techniques
- Generate synthetic data

Predictive Modeling

3



Processed Data



Conventional Machine Learning

- SVM
- Random Forest
- k-NN
- Logistic Regression



Prediction

- Disease risk
- Classification
- Outcome prediction



Deep Learning

- Neural Networks
- CNN
- RNN
- Transformer



Model Evaluation

- Accuracy
- Precision / Recall
- ROC-AUC
- Cross-validation



Optimized Model

Why is Bioinformatics important for CSE students?

6. You don't need to be a biology expert

- This is important 
As a CSE student:
- You learn **only the required biology**
- Focus is on **computation, algorithms, and analysis**
- Biology is treated like *domain knowledge*, not memorization

Bioinformatics in Machine Learning and AI

- ❑ Bioinformatics is the field where **biology + computer science + data analysis** are combined to study biological data such as DNA sequences, proteins, genes, and diseases.
- ❑ With the growth of Machine Learning (ML) and Artificial Intelligence (AI), bioinformatics has become one of the most important AI application areas.

Role of Machine Learning in Bioinformatics

□ Biological data are extremely large and complex. ML algorithms help computers identify patterns that humans cannot easily detect.

Examples:

a) Genomic Analysis

- DNA contains millions of genetic variations.
- ML models analyze genome sequences to:
 - identify disease-related mutations
 - predict genetic disorders
 - classify cancer types
- Example:
A deep learning model can analyze a patient's genome and predict the risk of diseases like cancer or diabetes.

Role of Machine Learning in Bioinformatics

b) Protein Structure Prediction

- Proteins are essential molecules in the body, but understanding their 3D structure is difficult.
- AI models can predict:
 - amino acid sequence → protein structure
 - protein function
 - protein interactions
- Example:

Google DeepMind developed AlphaFold, which uses deep learning to predict protein structures with high accuracy.

Role of Machine Learning in Bioinformatics

c) Drug Discovery Using AI

- Traditional drug discovery can take **10–15 years** and billions of dollars.
- AI helps by:
 - Finding disease-related targets
 - Designing new drug molecules
 - Predicting drug effectiveness
 - Predicting toxicity and side effects
- Workflow:
- **Disease data → AI model → Candidate drug → Laboratory testing → Clinical trial**

Role of Machine Learning in Bioinformatics

d) Disease Diagnosis

AI analyzes:

- medical images
- genetic data
- patient records
- biomarkers

Applications:

- cancer detection
- Alzheimer's disease prediction
- infectious disease identification

AI Techniques Used in Bioinformatics

□ Machine Learning Methods

1. Supervised Learning

- Uses labeled biological data.
- Example:
 - Predicting whether a gene mutation causes disease.
- Algorithms:
 - Support Vector Machine (SVM)
 - Random Forest
 - Logistic Regression

AI Techniques Used in Bioinformatics

2. Unsupervised Learning

- Finds hidden patterns in biological data.
- Example:
 - Grouping patients into different cancer subtypes.
- Algorithms:
 - K-means clustering
 - PCA

AI Techniques Used in Bioinformatics

3. Deep Learning

Uses neural networks to analyze complex biological data.

- Applications:
 - DNA sequence prediction
 - Protein folding
 - Medical image analysis
- Models:
 - CNN (Convolutional Neural Network)
 - RNN (Recurrent Neural Network)
 - Transformer models

Future of AI in Bioinformatics

Future developments include:

- **AI-designed medicines**
- **Personalized healthcare**
- **Faster vaccine development**
- **Digital twins of patients**
- **AI-assisted genetic engineering**
- **Early disease prediction**

Areas Where Bioinformatics is Applied!

- **Genomics:**
 - ★ Sequencing data analysis
 - ★ Genomic feature prediction
- **Proteomics:**
 - ★ Protein 3D structure modeling
 - ★ Drug design
- **System Biology:**
 - ★ Genome set enrichment
 - ★ Pathway analysis
- **Phenotype:**
 - ★ Image analysis
 - ★ Integration

What is the purpose of Bioinformatics?

- The work of bioinformatics is in the processing of large amount of data.
- The purpose of bioinformatics is to make sense of processes taking place inside organism or living systems.

Conventional Biology

- Focuses on **wet-lab experiments** (microscope, petri dish, DNA extraction, etc.).
- Studies living organisms directly (cells, tissues, plants, animals, microbes).
- **Example activities:**
 - Observing bacteria under a microscope.
 - Doing DNA sequencing in a lab.
 - Conducting breeding experiments with plants.

Bioinformatics

- Focuses on **computer-based analysis** of biological data.
- Uses algorithms, databases, and statistical tools to study biology.
- Handles **large-scale data** (like DNA sequences, protein structures, gene expression data).
- **Example activities:**
 - Comparing DNA sequences across species using software.
 - Predicting protein structures with algorithms.
 - Analyzing gene expression patterns in cancer research.

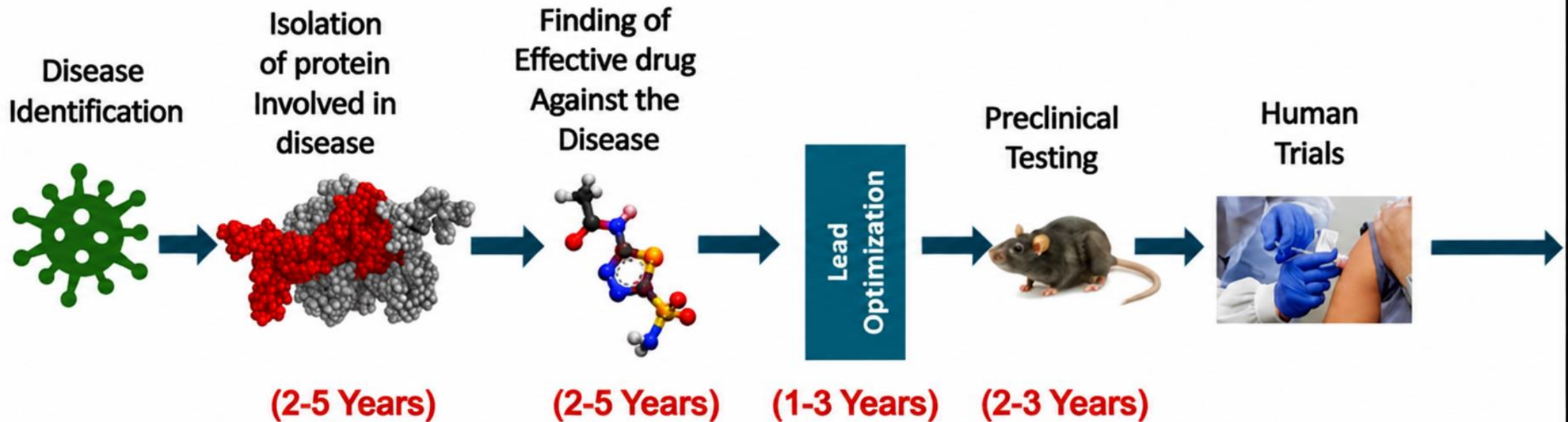
How is Bioinformatics Different from Conventional Biology

- **Drawbacks of conventional biology**
 - ★ **High resource requirement:** conventional research generally requires a sophisticated lab to work in and may also need expansive utilities and model organisms.
 - ★ **The experiments are normally long, delicate and sometimes risky to the researchers**

Let's take an example of drug discovery of a disease and see it in both ways: conventional biological approach and bioinformatic approach.

How is Bioinformatics Different from Conventional Biology

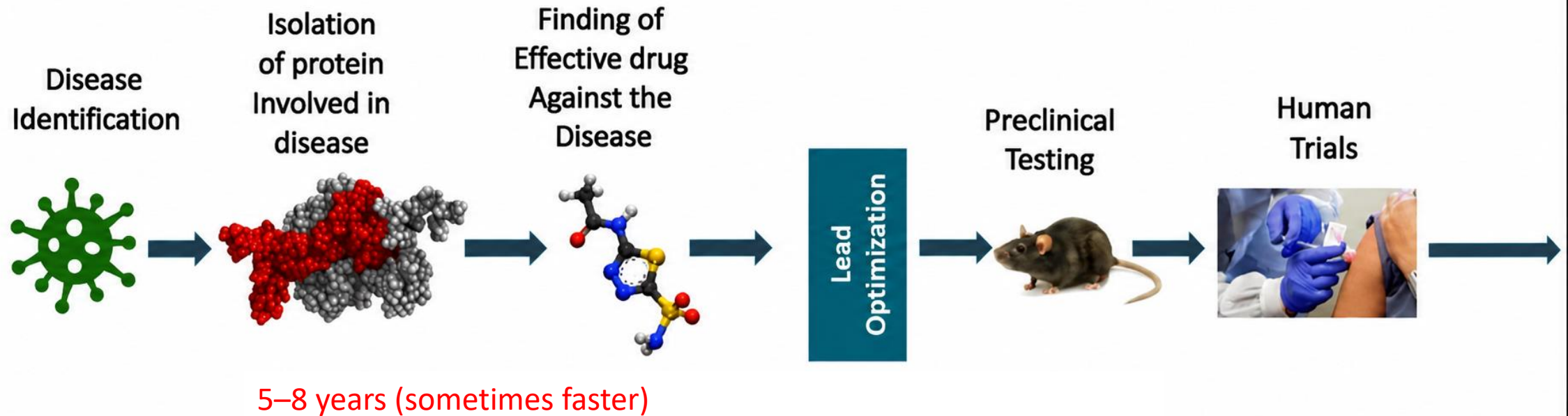
Conventional Drug Discovery

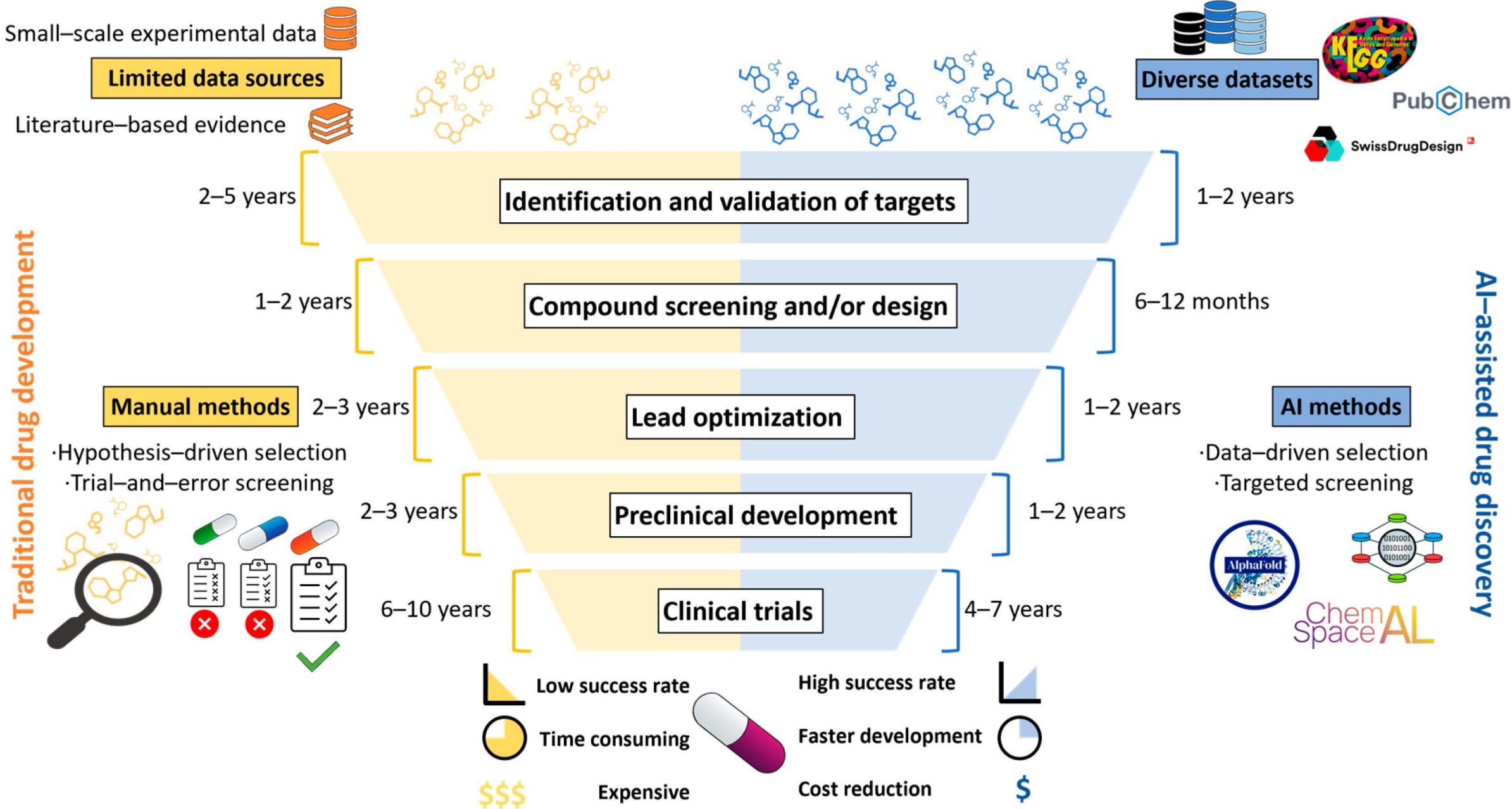


Conventional approach is lengthy, expansive and risky with high failure rate.

How is Bioinformatics Different from Conventional Biology

Drug Discovery Using Bioinformatics





Biological Database

These are the databases consisting of biological data like protein sequencing, molecular structure, DNA sequences, etc in an organized form.

Several computer tools are there to manipulate the biological data like an update, delete, insert, etc. Scientists, researchers from all over the world enter their experiment data and results in a biological database so that it is available to a wider audience.

Biological databases are free to use and contain a huge collection of a variety of biological data.

Uses of Biological Database

It helps the researchers to study the available data and form a new thesis, anti-virus, helpful bacteria, medicines, etc.

It helps scientists to understand the concepts of biological phenomena.

Secondary databases have become the molecular biologist's reference library over the past decade or so, providing a wealth of information on just about any gene or gene product that has been investigated by the research community.

The database acts as a storage of biological information.

It helps remove the redundancy of data.

There are basically three types of biological databases. They are: **Primary databases, secondary databases and composite databases.**

Primary Databases

- Stores **raw, experimental data** that comes directly from laboratory experiments without much interpretation.
 - **Nature:** Original, unprocessed, factual data.
 - **Examples:**
 - **GenBank** (DNA sequences – maintained by NCBI)
 - **EMBL-EBI** (European Molecular Biology Laboratory database)
 - **DDBJ** (DNA Data Bank of Japan)
 - **PDB** (Protein Data Bank – protein 3D structures)
- ✓ Example use: Uploading a newly sequenced gene into **GenBank** so others can access the raw sequence.

Types of Biological Database

Primary Databases

Accession Number in Bioinformatics

- A unique identifier assigned to each entry (sequence, structure, or dataset) when it is submitted to a **primary or secondary database**.
- No two records share the same accession number.
- Once assigned, it **never changes**, even if the entry is updated or corrected.

Purpose: Ensures reliable referencing, retrieval, and citation of biological data.

Examples:

- **GenBank (NCBI)**
 - Example accession: NM_000546 (human TP53 mRNA sequence).
- **PDB (Protein Data Bank)**
 - Example accession: 1A3N (a protein 3D structure).

Types of Biological Database

Secondary Databases

Stores **interpreted, curated, or derived data** that is analyzed and annotated from primary databases.

- **Nature:** Processed, predicted, or categorized information.

- **Examples:**

- **Pfam** (protein families and domains)
- **Prosite** (protein motifs and patterns)
- **InterPro** (integrated protein classification)
- **SCOP** (Structural Classification of Proteins)
- **Swiss-Prot (UniProtKB/Swiss-Prot)** (manually annotated protein sequences)

✓ Example use: Using **Pfam** to find which family a protein belongs to, based on sequences from primary databases.

Types of Biological Database

Composite Databases

A **composite database** is created by **integrating data from multiple primary databases** into a single, non-redundant, and more user-friendly resource.

- It collects raw sequence data (from primary sources).
- Removes duplicates and inconsistencies.
- Provides a consolidated dataset for easier searching and analysis.

Why needed?

Primary databases (like GenBank, EMBL, DDBJ) often contain duplicate entries because the same sequence might be submitted to more than one database. A composite database solves this by merging, cleaning, and unifying the data.

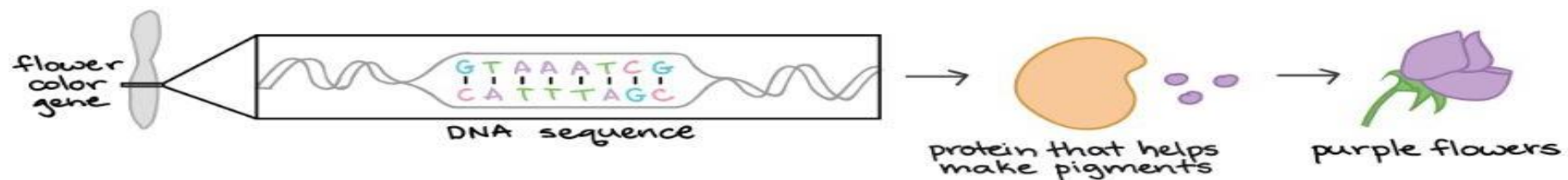
Examples:

Composite Databases -OWL, NRD and Swissport +TREMBL

Central Dogma of Molecular Biology

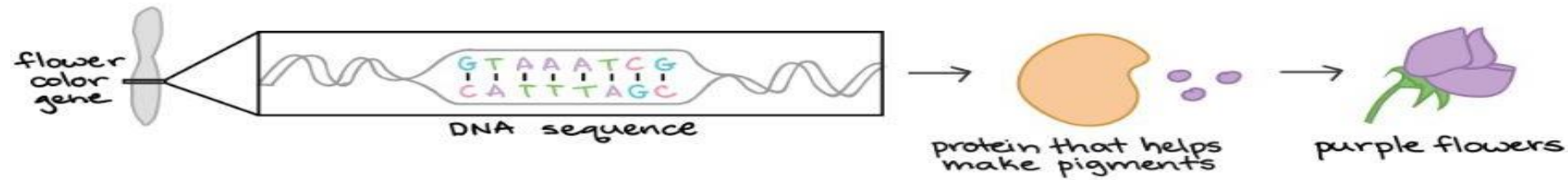
The bone, skin, and muscle that we see are made up of cells. And each of these cells contains many millions of proteins. As a matter of fact, **proteins are the key molecular building blocks for every organism on earth.** But how are these proteins made in a cell!!!

- The instructions for making proteins are written in a cell's DNA. A DNA is not just a long string of nucleotides but is divided up into functional units called genes.
- Each gene provides instructions for a functional product, that is, a molecule needed to perform a job in a cell. In many cases, the functional product of a gene is a protein.
- For example, Mendel's flower color gene provides instructions for a protein that helps make colored molecules (pigments) in flower petals.



- **Quick Example: Mendel's Flower Color Gene**

- DNA in the flower contains a gene for a pigment-making enzyme.
- Transcription → mRNA copy of that gene.
- Translation → protein (enzyme) that helps produce pigment molecules.
- Result → colored flower petals.



Central Dogma of Molecular Biology

The central dogma states that the pattern of information that occurs most frequently in our cells is:

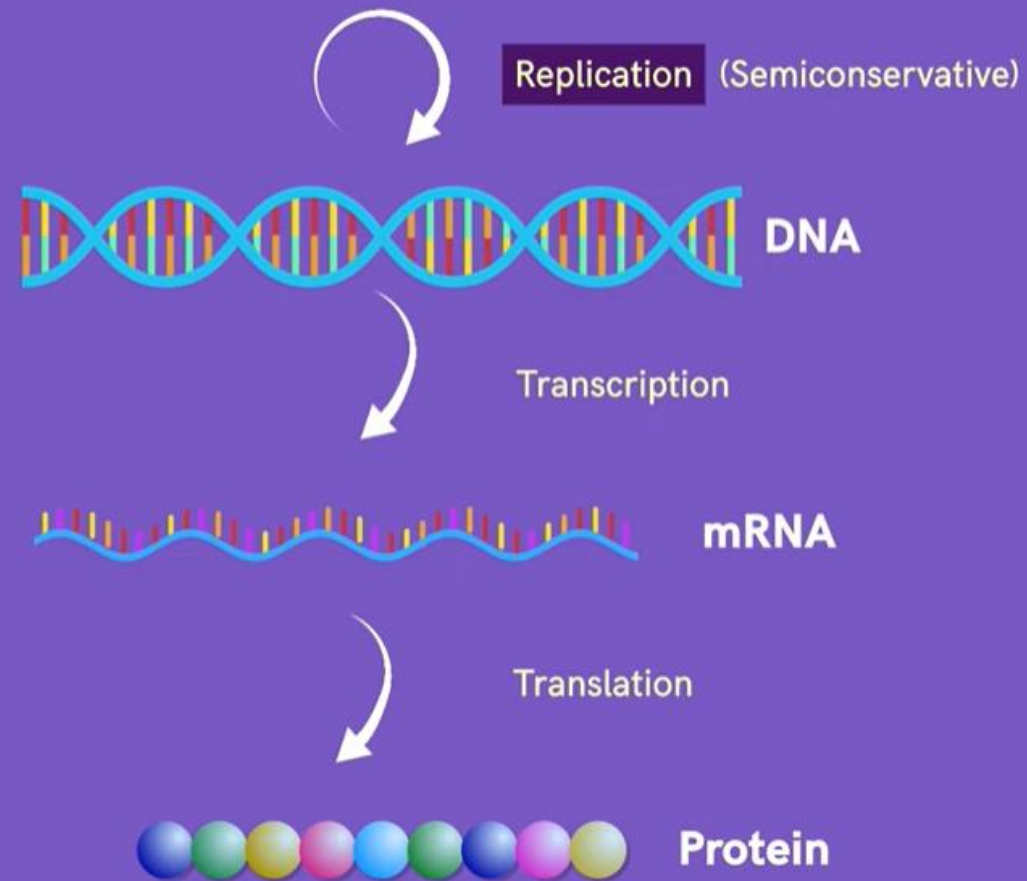
- From existing DNA to make new DNA (DNA replication)
- From DNA to make messenger RNA (Transcription)
- From mRNA to make new proteins (Translation)

In **transcription**, the DNA sequence of a gene is copied to make an RNA molecule. This step is called transcription because it involves rewriting, or transcribing, the DNA sequence in a similar RNA "alphabet." In eukaryotes, the RNA molecule must undergo processing to become a mature messenger RNA (mRNA).

•**DNA alphabet = A, T, C, G**

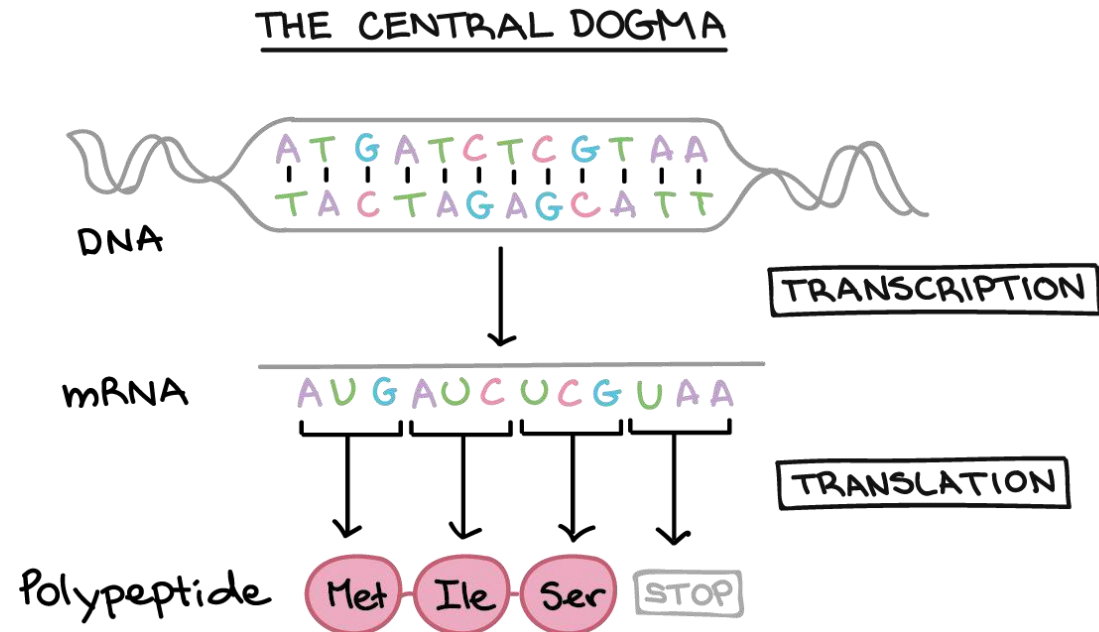
•**RNA alphabet = A, U, C, G**

Central Dogma



Central Dogma of Molecular Biology

In **translation**, the sequence of the mRNA is decoded to specify the amino acid sequence of a polypeptide. The name translation reflects that the nucleotide sequence of the mRNA sequence must be translated into the completely different "language" of amino acids.



Thus, during expression of a protein-coding gene, information flows from DNA → RNA → protein. This directional flow of information is known as the central dogma of molecular biology. Non-protein-coding genes (genes that specify functional RNAs) are still transcribed to produce an RNA, but this RNA is not translated into a polypeptide. For either type of gene, the process of going from DNA to a functional product is known as gene expression.

The Genetic Code

The genetic code is the **set of rules** that defines how the sequence of nucleotides in mRNA is translated into a sequence of amino acids in a protein.

Codons

- A codon is a sequence of three nucleotides in mRNA.
- Each codon specifies one amino acid (or a stop signal).
- Since there are 4 nucleotides (A, U, C, G), there are $4^3 = 64$ possible codons.
- 20 commonly found in proteins

Start and Stop Codons

1. Start codon

- AUG → signals **start of translation**
- Also codes for **methionine** (first amino acid in most proteins)

2. Stop codons

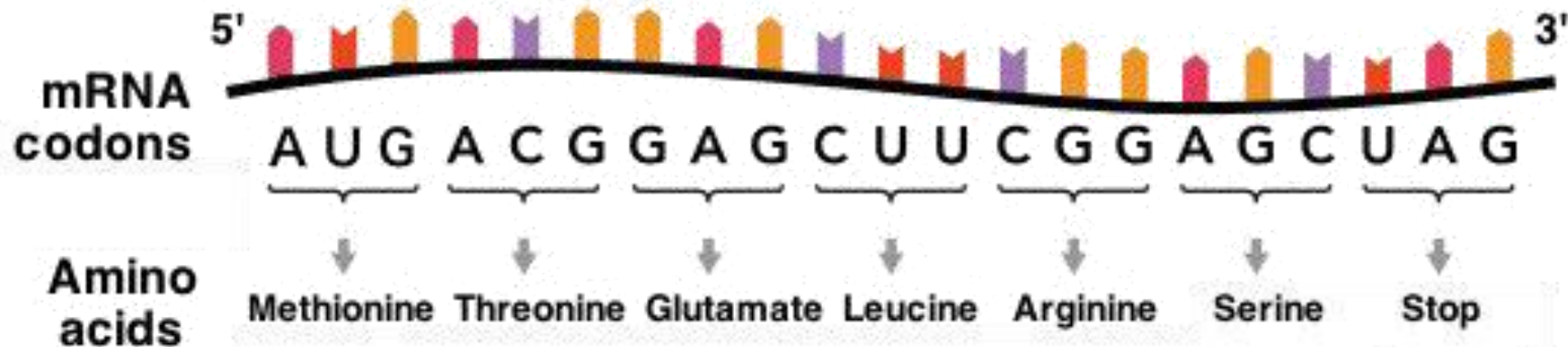
- UAA, UAG, UGA → **do not code for any amino acid**
- Signal the **end of the polypeptide chain**

The Genetic Code

		Second letter				
		U	C	A	G	
First letter	U	UUU } Phe UUC } UUA } Leu UUG }	UCU } UCC } Ser UCA } UCG }	UAU } Tyr UAC } UAA Stop UAG Stop	UGU } Cys UGC } UGA Stop UGG Trp	U C A G
	C	CUU } CUC } Leu CUA } CUG }	CCU } CCC } Pro CCA } CCG }	CAU } His CAC } CAA } Gln CAG }	CGU } CGC } Arg CGA } CGG }	U C A G
	A	AUU } AUC } Ile AUA } AUG Met	ACU } ACC } Thr ACA } ACG }	AAU } Asn AAC } AAA } Lys AAG }	AGU } Ser AGC } AGA } Arg AGG }	U C A G
	G	GUU } GUC } Val GUA } GUG }	GCU } GCC } Ala GCA } GCG }	GAU } Asp GAC } GAA } Glu GAG }	GGU } GGC } Gly GGA } GGG }	U C A G

The Genetic Code

There are three more codons that do not specify amino acids. These stop codons, UAA, UAG, and UGA, tell the cell when a polypeptide is complete. All together, this collection of codon-amino acid relationships is called the genetic code, because it tells cells “decode” an mRNA into a chain of amino acids.

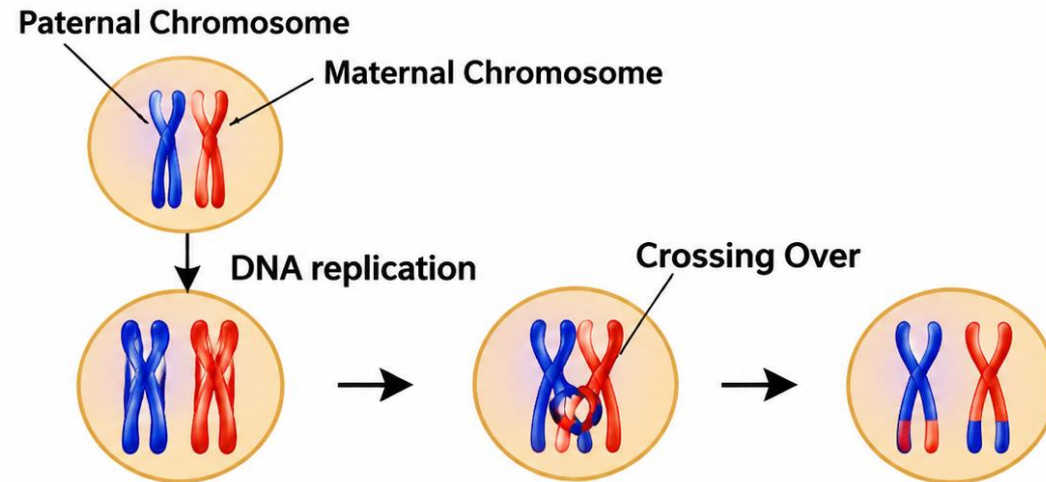


Evidence of Evolution

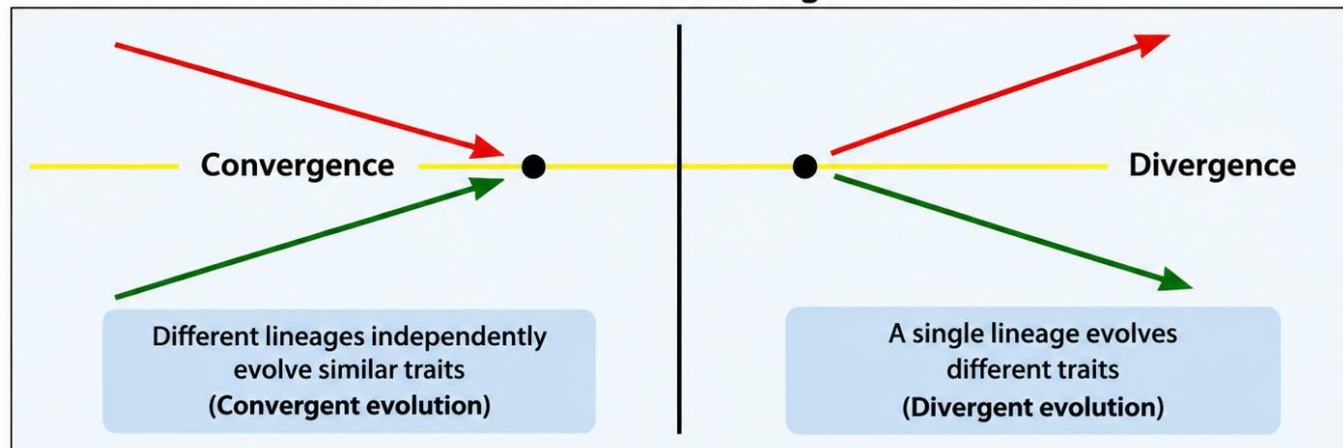
- Evolution is the process by which species **change over time**, giving rise to new forms while sharing common ancestry with earlier organisms.
- **Four sources of evidences:**
 - I. **Fossil Record**
 - II. **Geographical Distribution of Related Species**
 - III. **Homologous body structures**
 - IV. **Similarities in embryology**

Evidence of Evolution

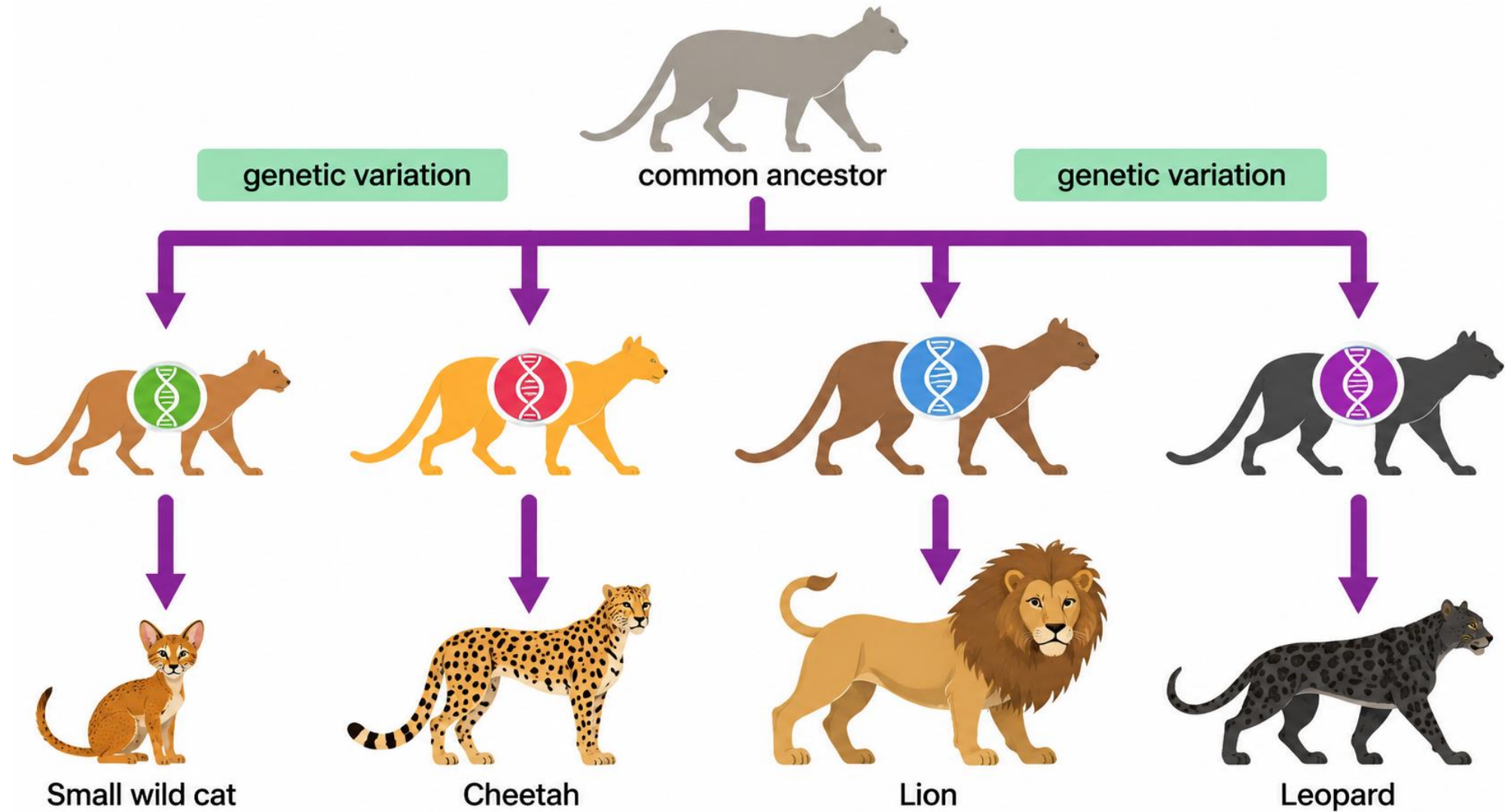
- The mechanism through which evolution occur include:
 - A. Natural Selection
 - B. Mutation
 - C. Sexual Recombination
 - D. Divergence
 - E. Convergence



Recombination between 2 homologous chromosomes



Evidence of Evolution



Types of Evolution

- **Evolution can be of several types:**
 1. **Divergent evolution**
 2. **Convergent evolution**
- **Speciation: Divergent evolution may lead to completely new species sometimes. This is known as speciation.** Speciation is the evolutionary process by which new species arise from existing ones. It often occurs when populations of the same species become **isolated** from one another and undergo different evolutionary pressures over time.

Divergent Evolution

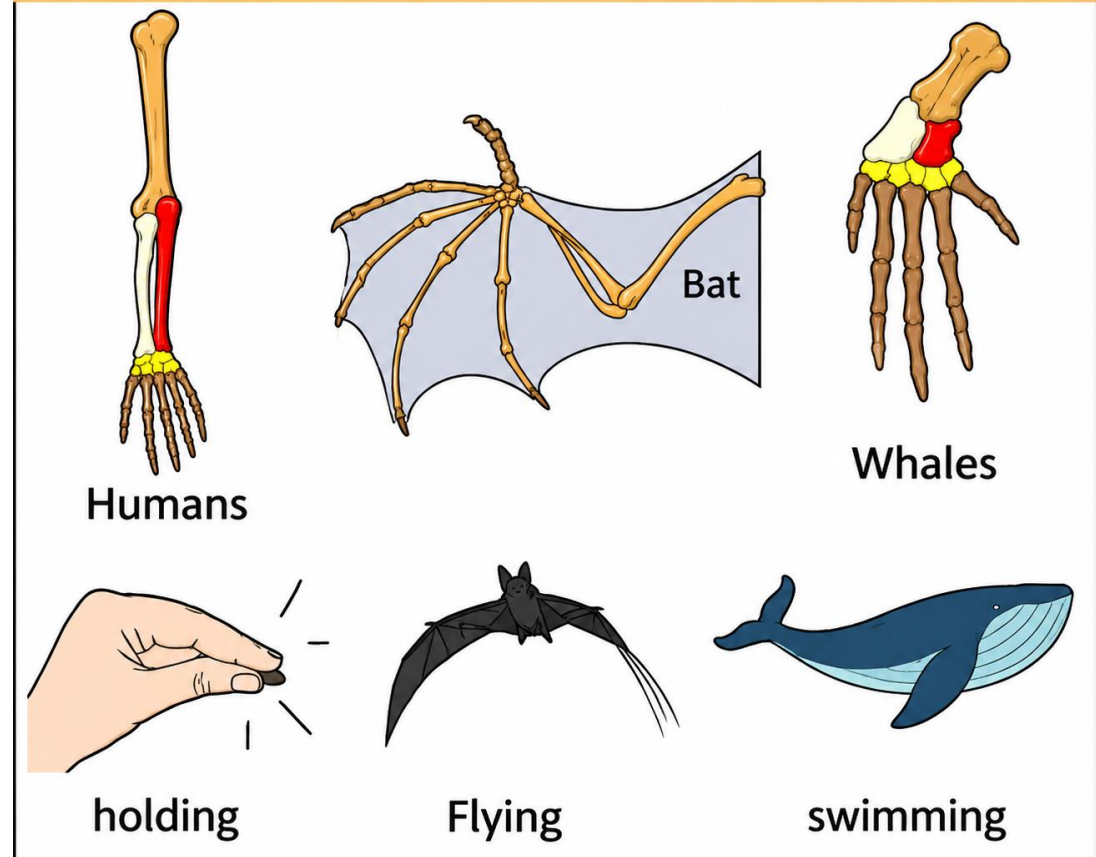
When two or more species evolve from a **common ancestor** but adapt to **different environments**.

- **Result:** Structures remain **homologous** (same origin, different functions).
- **Example:** Human arm, bat wing, whale flipper.

Types of Evolution (Divergent Evolution)

- Here, human, bat and whales are mammals.
- **Homologous organs:** organs with similar structure (skeletal structure or anatomy) and origin, but with different function in different organisms.
- Homologous organs are developed in related animals.
- Homologous organs are inherited from a common ancestor.

Pentadactyl limbs from amphibians to mammals



Types of Evolution (Convergent Evolution)

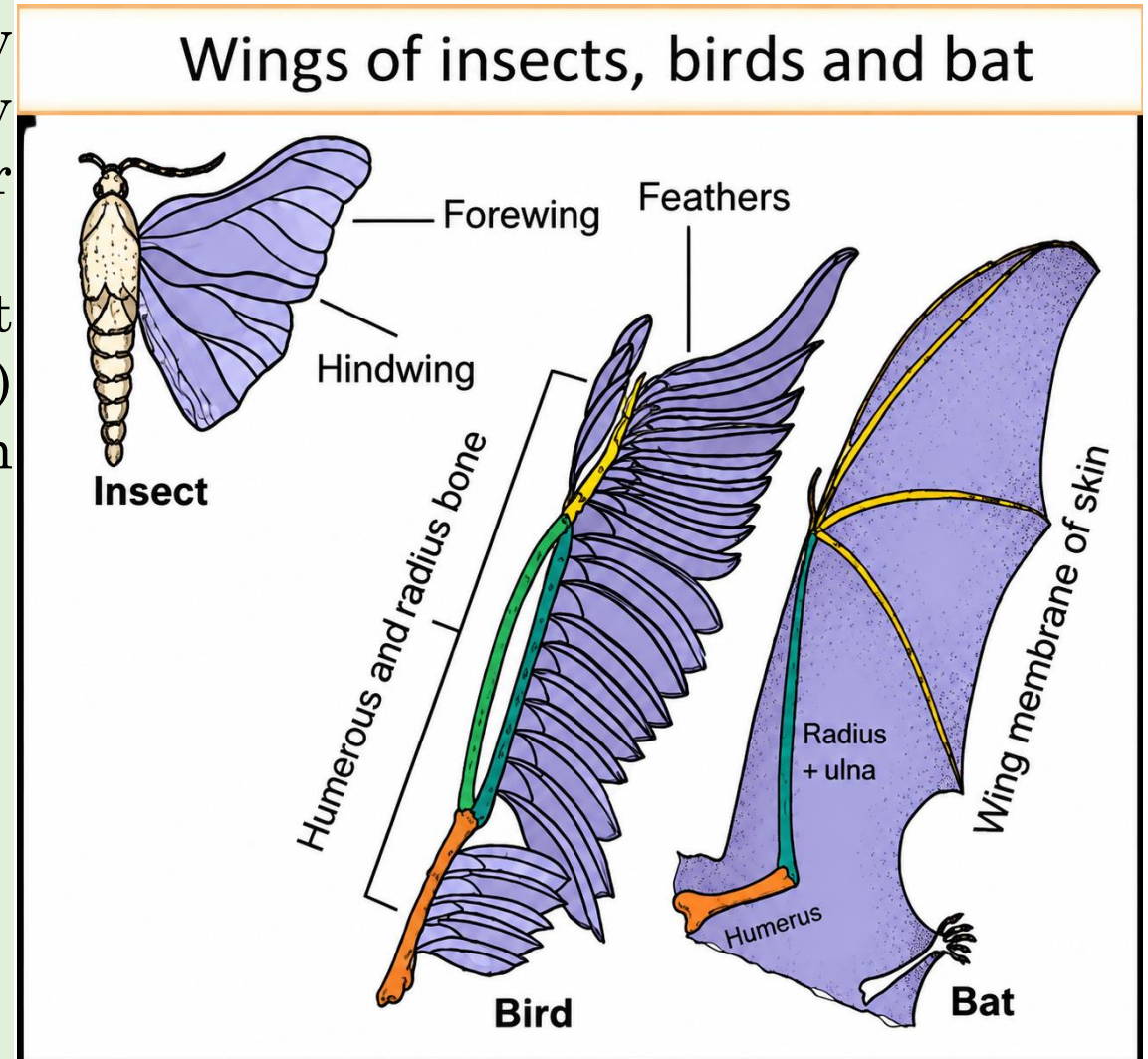
- **Convergent evolution:** It is the process by which unrelated groups independently evolve similar traits to adapt to similar environment.
- **Analogous organs:** organs with different structure (skeletal structure or anatomy) and origin, but with similar function in different organisms.

Key Features:

- Different anatomy, **same function**
- Found in **unrelated animals**
- Evolved due to **similar environments or lifestyles**
- Evidence of **convergent evolution**

Example:

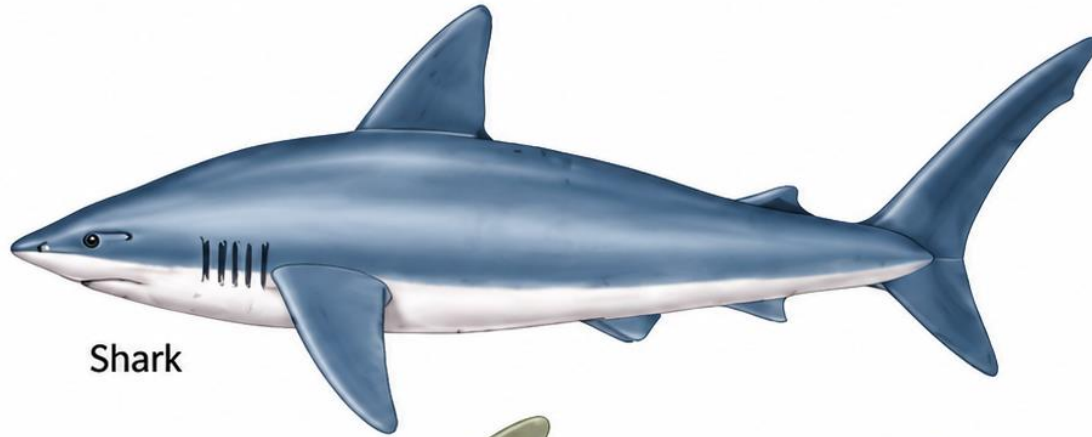
- Wings of bats (mammals) and insects (arthropods).
- Flippers of dolphins (mammals) and sharks (fish).



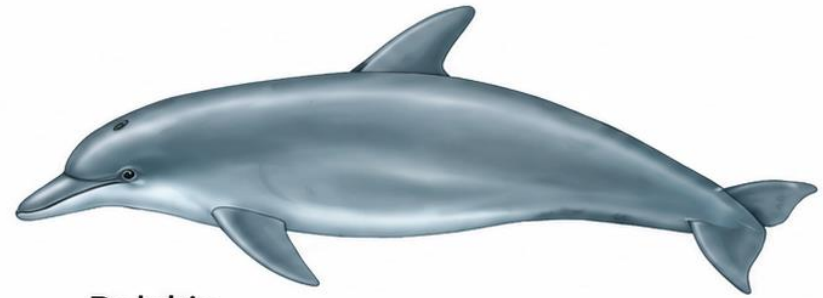
Types of Evolution (Convergent Evolution)

Convergent Evolution: Streamlining

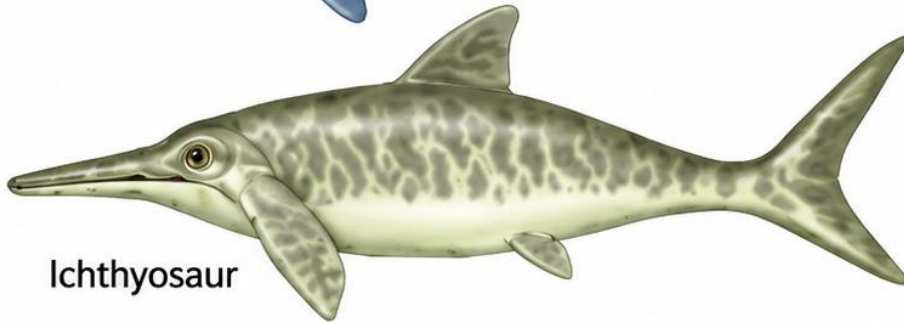
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Shark



Dolphin



Ichthyosaur



Penguin

Convergent evolution is the process by which unrelated species evolve similar physical characteristics because they have similar lifestyles